American Computer Science League

2021-2022 ● Contest 3: Short Problem Solutions ● Elementary Division

1. Boolean Algebra Remember that the order of Boolean operators is NOT, AND, then OR. If x = 2, NOT (2=10) AND (2≥6) OR (2=5) = (T and F) or F = FALSE. If x = 5, NOT (5=10) AND (5≥6) OR (5=5) = (T and F) or T = TRUE. If x = 6, NOT (6=10) AND (6≥6) OR (6=5) = (T and T) or F = TRUE. If x = 10, NOT (10=10) AND (10 ≥ 6) OR (10=5) = (F and T) or F = FALSE. If x = 13, NOT (13=10) AND (13 ≥ 6) OR (13=5) = (T and T) or F = TRUE. Therefore, the answer is that there are 3 of them that make it TRUE.	C. 3
2. Boolean Algebra The ordered pair(s) that make ~(A~B + ~A~B) FALSE, make A~B + ~A~B TRUE. Therefore, the two ordered pairs are A=1, B=0 or A=0, B=0. That gives (1,0) and (0,0).	E. (1,0), (0,0)
3. Boolean Algebra B(A~B + A) = AB~B + AB = AB A~B + A = A(~B + 1) = A Therefore, the two that are equivalent are A and D.	A. a, d
4. Boolean Algebra A~B(A + ~B) + ~AB(~A + B) = A~BA + A~B~B + ~AB~A + ~ABB = A~B + A~B + ~AB + ~AB = A~B + ~AB	B. A~B + ~AB
5. Boolean Algebra A tautology is always TRUE. Each of the expressions simplifies: A~B+~B+A = ~B(A+1)+A = ~B+A is TRUE for (0,0), (1,1), (1,0). ~(A+B) + (A+B) = ~A~B + A + B is TRUE for (0,0), (1,0), (1,1), (0,1). ~A(A~B) + ~B(~AB) = ~AA~B + ~AB~B = 0+ 0=0 is never TRUE. AB+~AB+A = B(A+~A)+A = B(1)+A = B+A is TRUE for (1,1), (0,1), (1,0).	B. ~(A + B) + (A + B)